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14.2.26 Assignment -3

A3.1 (a) $S_{vi} = \frac{V_p^2}{2} \delta(f-f_0)$; $S_{vna} = 4\pi f R$

→ $v_o = \frac{v_i}{1 + sRC}$ ⇒ $S_{v_o}(f) = \frac{V_p^2 \delta(f-f_0)}{2[1 + (2\pi f RC)^2]}$

• $\sigma_{v_o}^2 = \text{MS value} = \int_{-\infty}^{\infty} S_{v_o}(f) df$

$\sigma_{v_o}^2 = \frac{V_p^2}{2[1 + (2\pi f RC)^2]} \int_{-\infty}^{\infty} \delta(f-f_0) df = \frac{V_p^2}{4}$

→ $v_{no} = \frac{v_{na}}{1 + sRC}$ ⇒ $S_{v_{no}}(f) = \frac{4\pi f R}{1 + (2\pi f RC)^2}$

∴ $\sigma_{v_{no}}^2 = \text{MS value} = \frac{4\pi f R}{2\pi RC} \int_{-\infty}^{\infty} \frac{d(2\pi f RC)}{1 + (2\pi f RC)^2}$

$\sigma_{v_{no}}^2 = \frac{4\pi f R \times \pi}{2\pi RC \cdot 2} = \frac{\pi f}{C}$

→ $S_{Na} = \frac{\sigma_{v_o}^2}{\sigma_{v_{no}}^2} = \frac{V_p^2 \times C}{4 \times \pi f}$

→ $x(t) = v(t) \times i(t)$; $v(t) = V_p \cos \omega_0 t$

$i(t) = \frac{V_p}{\sqrt{R^2 + X_C^2}} \cos(\omega_0 t + \phi)$; $\phi = \tan^{-1}\left(\frac{X_C}{R}\right)$

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$$x_c = \frac{1}{\omega_0 C} \Rightarrow x_c = R \Rightarrow \phi = 45^\circ$$

$$\therefore i(t) = \frac{V_R}{R\sqrt{2}} \cos(\omega_0 t + 45^\circ)$$

$$p(t) = \frac{V_R^2}{R\sqrt{2}} \cos \omega_0 t \cos(\omega_0 t + 45^\circ)$$

$$P_{avg} = \frac{1}{T} \int_0^T p(t) dt \quad ; \quad \omega_0 = \frac{2\pi}{T}$$

$$= \frac{V_R^2}{R\sqrt{2}} \left[\int_0^T \cos(2\omega_0 t + 45^\circ) dt + \int_0^T \cos 45^\circ dt \right]$$

$$P_{avg} = \frac{V_R^2}{4R}$$

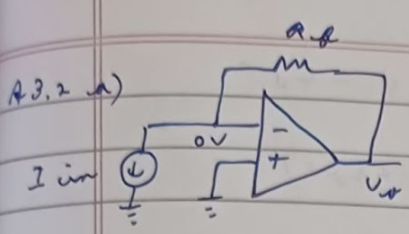
b) $P_{avg} = (2\pi kT) \times SNR \times f_B \quad ; \quad f_B = \frac{1}{2\pi RC}$

- If we need either high SNR or high f_B , we would need to dissipate more power.
- If we only have a fixed power, then any increase in SNR would be at the cost of bandwidth (and vice versa).

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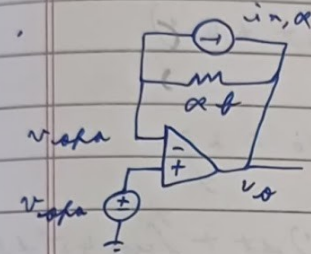
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Q3.2 a)



$$\frac{v_o - 0}{R_f} = I_{in} \Rightarrow \frac{v_o}{R_f} = I_{in}$$

$$\therefore S_{V_{out}} = R_f^2 \times S_{I_{n, in}}$$



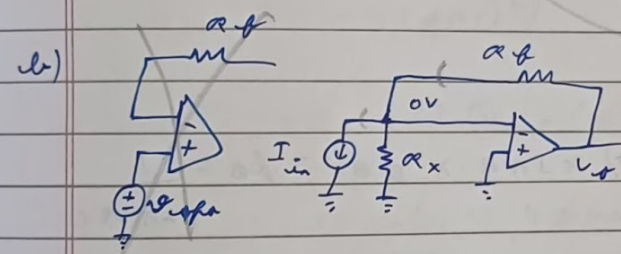
$$\frac{v_o - v_{o, Rf}}{R_f} = i_{n, Rf}$$

$$\therefore S_{v_o} = R_f^2 S_{i_{n, Rf}} + S_{v_{o, Rf}}$$

$$\therefore S_{I_{n, in}} = S_{i_{n, Rf}} + \frac{S_{v_{o, Rf}}}{R_f^2}$$

$$= \frac{1}{R_f} + \frac{S_{v_{o, Rf}}}{R_f^2}$$

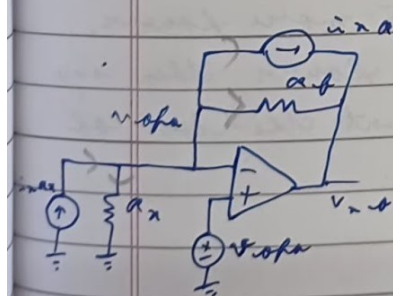
b)



$$\frac{v_o - 0}{R_f} = I_{in}$$

$$\therefore \frac{v_o}{I_{in}} = R_f$$

$$\therefore S_{V_{out}} = R_f^2 \times S_{I_{n, in}}$$



$$i_{n, Rx} = \frac{v_{o, Rx}}{R_x} + i_{n, Rf} + \frac{(v_{o, Rx} - v_o)}{R_f}$$

$$\therefore v_o = R_f (i_{n, Rf} - i_{n, Rx}) + R_f v_{o, Rx} \left(\frac{1}{R_f} + \frac{1}{R_x} \right)$$

$$\therefore S_{V_{out}} = R_f^2 (S_{i_{n, Rf}} + S_{i_{n, Rx}}) + R_f^2 \left(\frac{1}{R_f} + \frac{1}{R_x} \right)^2 S_{v_{o, Rx}}$$

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$$\therefore S_{I_{in}, in} = 4kT \left(\frac{1}{\alpha_f} + \frac{1}{\alpha_x} \right) + S_{V_{ofa}} \left(\frac{1}{\alpha_f} + \frac{1}{\alpha_x} \right)^2$$

c)

$$\frac{v_x \times k}{\alpha_f} = I_{in}$$

$$\frac{v_o}{\alpha} = -\frac{v_x \times k}{\alpha}$$

$$\therefore \frac{v_o}{I_{in}} = -\alpha_f \Rightarrow S_{V_{ofa}} = \alpha_f^2 \times S_{I_{in}, in}$$

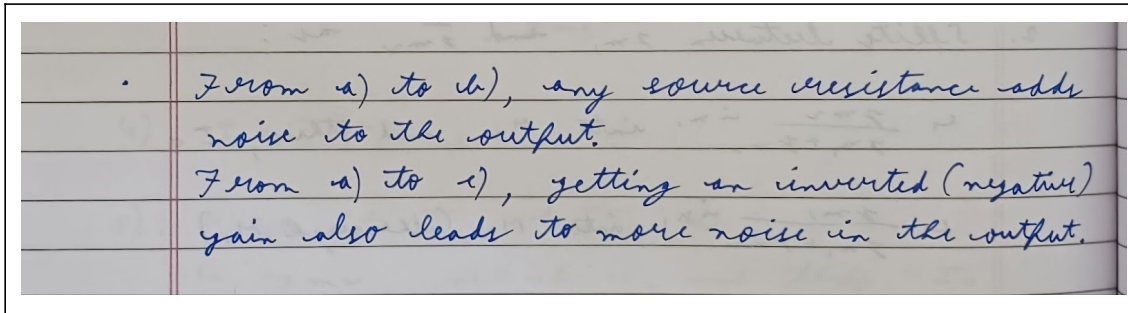
$$\frac{(v_x - v'_{ofa})k}{\alpha_f} = \frac{i_{na} \alpha_f / k}{\alpha_f}$$

$$\left. \begin{aligned} & \frac{(v_x - v'_{ofa})k}{\alpha_f} = \frac{i_{na} \alpha_f / k}{\alpha_f} \\ & \frac{(v_x - v'_{ofa})k}{\alpha_f} = \frac{v'_{ofa} - v_o}{\alpha} + i_{na} \end{aligned} \right\}$$

$$\therefore v_o = v'_{ofa} (k+1) - v_{ofa} \times k - \alpha_f i_{na} \alpha_f / k + \alpha (i_{na} - i_{na} / k)$$

$$S_{I_{in}, in} = \frac{S_{V_{ofa}}}{\alpha_f^2} = \frac{S_{V_{ofa}}}{\alpha_f^2} (2k^2 + 2k + 1) + \frac{4kT}{\alpha_f^2} [k\alpha_f + (k+1)\alpha]$$

* $S_{V_{ofa}, in} = 0$ ($\because 2i_{in} = 0$) in all 3

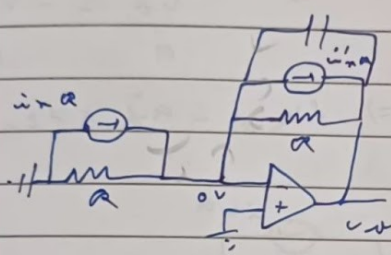


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A3.3 a) $\frac{v_i}{R} = \frac{-v_o}{\alpha // \frac{1}{sC}} \Rightarrow \frac{v_o}{v_i} = \frac{-1}{1 + s\alpha C}$

$\therefore \omega_{-3dB} = 10^4 = \frac{1}{\alpha C} = \frac{1}{\alpha \times 10^{-12}} \Rightarrow \alpha = 100 \text{ M}\Omega$



$i_{in} + \frac{v_o}{\alpha} + v_o sC = i_{in}$

$\therefore v_o = \alpha (i_{in} - i_{in})$

$\therefore s_{v_{no}} = \frac{\alpha^2 \times 2s i_{in}}{1 + (2\pi f \alpha C)^2} = \frac{8\pi \alpha}{1 + (2\pi f \alpha C)^2}$

$\therefore v_{no}^2 = \int_0^\infty s_{v_{no}} df = \frac{2\pi \alpha}{C}$

$\therefore \text{rms value} = \sqrt{\frac{2 \times 4 \times 10^{-21}}{10^{-12}}} = 89.44 \mu\text{V}$

$s_{v_{no}} = \frac{s_{v_{in}}}{1 + (2\pi f \alpha C)^2} = \frac{8\pi \alpha}{1 + (2\pi f \alpha C)^2}$

$\therefore s_{v_{in}} = 8\pi \alpha = 32 \times 10^{-13} \text{ V}^2/\text{Hz}$

Total resistance = $2\alpha = 200 \text{ M}\Omega$

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b)

$$\frac{v_i - v_1}{\alpha_1} = \frac{v_1}{\alpha_2} + \frac{v_1}{\alpha_1}$$

$$\therefore v_1 = \frac{\alpha_2 v_i}{\alpha_1 + 2\alpha_2} \quad (1)$$

$$\frac{v_1}{\alpha_1} + v_o(s) + \frac{v_2}{\alpha_1} = 0 \quad (2)$$

$$\frac{v_o - v_2}{\alpha_1} = \frac{v_2}{\alpha_2} + \frac{v_2}{\alpha_1} \Rightarrow v_2 = \frac{\alpha_2 v_o}{\alpha_1 + 2\alpha_2} \quad (3)$$

$$\therefore \frac{v_o}{v_i} = \frac{-1}{1 + s\alpha_1\left(2 + \frac{\alpha_1}{\alpha_2}\right)} = \frac{-1}{1 + s\alpha_1}$$

$$\therefore 10^8 = \alpha_1 \left(2 + \frac{\alpha_1}{\alpha_2}\right) \Rightarrow \alpha_1^2 + 2\alpha_1\alpha_2 - 10^8\alpha_2 = 0$$

$$\therefore \alpha_1 = -\alpha_2 \pm \sqrt{\alpha_2^2 + \alpha_2 \times 10^8} \quad (4)$$

$$2\alpha_1 + \alpha_2 \leq 10^6 \Omega \quad (5)$$

$$\therefore -\alpha_2 + \sqrt{4\alpha_2^2 + 4\alpha_2 \times 10^8} \leq 10^6$$

$$\therefore 3\alpha_2^2 + \alpha_2 \times 398 \times 10^6 - 10^{12} \leq 0$$

$$\therefore \alpha_2 < \frac{10^6}{3} \left(-398 + \sqrt{398^2 + 12} \right) = 2.512 \text{ k}\Omega$$

$$\therefore \alpha_1 = 498.692 \text{ k}\Omega$$

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c)

$$\therefore i_{n\alpha_{1,1}} + i_{n\alpha_{2,1}} = \frac{2V_1}{\alpha_1} + \frac{V_1}{\alpha_2} + i_{n\alpha_{1,2}}$$

$$i_{n\alpha_{1,2}} + \frac{V_1}{\alpha_1} + \frac{V_o(s)}{\alpha_1} + \frac{V_2}{\alpha_1} = i_{n\alpha_{1,3}}$$

$$i_{n\alpha_{1,3}} + i_{n\alpha_{2,2}} + \frac{V_o - V_2}{\alpha_1} = \frac{V_2}{\alpha_1} + \frac{V_2}{\alpha_2} + i_{n\alpha_{1,4}}$$

$$\therefore V_1 = \frac{\alpha_1 \alpha_2}{2\alpha_2 + \alpha_1} (i_{n\alpha_{1,1}} + i_{n\alpha_{2,1}} - i_{n\alpha_{1,2}})$$

$$V_o = \alpha_1 \left[\frac{\alpha_2 i_{n\alpha_{1,1}} + (\alpha_1 + \alpha_2) i_{n\alpha_{1,2}} - (\alpha_1 + \alpha_2) i_{n\alpha_{1,3}} - \alpha_2 i_{n\alpha_{1,4}} + \alpha_2 i_{n\alpha_{2,1}} + \alpha_2 i_{n\alpha_{2,2}}}{\alpha_2 + s(\alpha_1 (2\alpha_2 + \alpha_1))} \right]$$

$$\therefore S_{V_{no}} = \frac{8kT\alpha_1 [\alpha_1^2 + 3\alpha_1\alpha_2 + 2\alpha_2^2]}{\alpha_1^2 + [2\pi f(\alpha_1 (2\alpha_2 + \alpha_1))]^2}$$

$$\therefore r_{V_{no}}^2 = \frac{8kT\alpha_1 (\alpha_1 + 2\alpha_2) (\alpha_1 + \alpha_2)}{\alpha_1^2 \times 2\pi(\alpha_1 (2\alpha_2 + \alpha_1))} \int_0^\infty \frac{d[2\pi f \frac{\alpha_1}{\alpha_2} (2\alpha_2 + \alpha_1)]}{1 + [2\pi f \frac{\alpha_1}{\alpha_2} (2\alpha_2 + \alpha_1)]^2}$$

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$$\therefore r_{v_{n\theta}}^2 = \frac{2kT}{C} \left(\frac{\alpha_1 + \alpha_2}{\alpha_2} \right)$$

$$\therefore \text{RMS } v_{n\theta} = \sqrt{\frac{2kT}{C}} \sqrt{\frac{1 + \frac{498692}{2512}}{2512}} = 1.283 \text{ mV}$$

$$\rightarrow S_{v_{n\theta}} = \frac{S_{v_{n, in}}}{1 + (2\pi f(R))^2} = \frac{\alpha_v^2 S_{v_{n, in}}}{\alpha_v^2 + [2\pi f(R, (2\alpha_v + \alpha_1))]^2}$$

$$\therefore S_{v_{n, in}} = \frac{8kT\alpha_1}{\alpha_v} \frac{(\alpha_1 + \alpha_v)(2\alpha_v + \alpha_1)}{\alpha_v}$$

$$= 8kT \left(1 + \frac{\alpha_1}{\alpha_v}\right) \times \alpha_1 \left(2 + \frac{\alpha_1}{\alpha_v}\right)$$

$$= 8kT \alpha \left(1 + \frac{498692}{2512}\right) = 63.848 \times 10^{-11} \text{ V}^2/\text{Hz}$$

• While the total resistance needed on the chip can be decreased by using such techniques, the noise increases significantly.

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A 3.4 a)

We will now solve for each of the 6 noise currents individually and combine them with superposition:-

1. I_o of $\frac{1}{g_{m3}}$ up, ∞ down
 $\therefore g_{m3} v_y = -v_x, \quad \left\{ I_o = -g_{m4} v_y = \frac{g_{m4}}{g_{m3}} v_x \right.$
2. Split between $\frac{1}{g_{m1}}$ and $\frac{1}{g_{m2}}$ as:
 $\hookrightarrow \frac{g_{m2}}{g_{m1} + g_{m2}} v_x$ into M_2 and then, I_o
 $\hookrightarrow \frac{g_{m1}}{g_{m1} + g_{m2}} v_x$ into M_3 (through M_1)
 $\therefore I_o = \frac{-g_{m4}}{g_{m3}} \left(\frac{g_{m1}}{g_{m1} + g_{m2}} \right) v_x$

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combining the 3:- $I_0 = i\pi_1 \left(\frac{g_{m1}}{g_{m3}} \right) \left(\frac{g_{m3} + g_{m4}}{g_{m1} + g_{m2}} \right)$

2)

1. $I_0 = -i\pi_2$ directly.
2. Splits between $\frac{1}{g_{m1}}$ and $\frac{1}{g_{m2}}$ as:
 - $\hookrightarrow \frac{g_{m2}}{g_{m1} + g_{m2}} i\pi_2$ into M_2 and thus I_0
 - $\hookrightarrow \frac{g_{m1}}{g_{m1} + g_{m2}} i\pi_2$ into M_3 (via M_1)
$$\therefore I_0 = -\frac{g_{m4}}{g_{m3}} \left(\frac{g_{m1}}{g_{m1} + g_{m2}} \right) i\pi_2$$

combining the 3, $I_0 = -i\pi_2 \left(\frac{g_{m1}}{g_{m3}} \right) \left(\frac{g_{m3} + g_{m4}}{g_{m1} + g_{m2}} \right)$

3)

1. $I_0 = 0$
2. $2i\pi_3 = \frac{1}{g_{m3}}$ up, ∞ down

$\therefore i\pi_3$ into M_3 and gets mirrored into M_4

$\therefore I_0 = -\frac{g_{m4}}{g_{m3}} i\pi_3$

4) $I_0 = i\pi_4$

5) Splits into $\frac{1}{g_{m1}}$ and $\frac{1}{g_{m2}}$ as:

- $\hookrightarrow \frac{g_{m2}}{g_{m1} + g_{m2}}$ out of M_2 and thus $-I_0$

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$\hookrightarrow \frac{I_{m1}}{I_{m1} + I_{m2}}$ out of M_3 (via M_1) + mirrored by M_4
 $\therefore I_0 = +I_{m4} \left(\frac{I_{m1}}{I_{m1} + I_{m2}} \right) i_{x0}$

combining the 2, $I_0 = \frac{(I_{m1} I_{m4} - I_{m2} I_{m3})}{I_{m3} (I_{m1} + I_{m2})} i_{x0}$

6) $I_{m00} v_h = -i_{x00} \Rightarrow \frac{I_{m0}}{I_{m00}} i_{x00}$ from M_0 (upwards)
 $\therefore I_0 = \frac{-I_{m0} i_{x00} (I_{m1} I_{m4} - I_{m2} I_{m3})}{I_{m00} I_{m3} (I_{m1} + I_{m2})}$

$I_0 = i_{x1} \left(\frac{I_{m2}}{I_{m3}} \right) \left(\frac{I_{m3} + I_{m4}}{I_{m1} + I_{m2}} \right) \oplus i_{x2} \left(\frac{I_{m1}}{I_{m3}} \right) \left(\frac{I_{m3} + I_{m4}}{I_{m1} + I_{m2}} \right)$
 $- i_{x3} \left(\frac{I_{m4}}{I_{m3}} \right) + i_{x4} + \frac{i_{x0} (I_{m1} I_{m4} - I_{m2} I_{m3})}{I_{m3} (I_{m1} + I_{m2})}$
 $- i_{x00} \left(\frac{I_{m0}}{I_{m00}} \right) \frac{(I_{m1} I_{m4} - I_{m2} I_{m3})}{I_{m3} (I_{m1} + I_{m2})}$

This can also be arrived at by solving
 1. $i_{x3} + I_{m1} v_x = I_{m3} v_f + i_{x1}$
 2. $I_0 + I_{m4} v_f + i_{x2} = i_{x4} + I_{m2} v_x$
 3. $I_{m1} v_x + i_{x0} + I_{m2} v_x + I_{m0} v_h = i_{x2} + i_{x1}$
 4. $I_{m00} v_h = -i_{x00}$

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* Assuming $g_{m1} = g_{m2} = g_{mn}, g_{m3} = g_{m4} = g_{mp}$

$$I_0 = i_{n1} - i_{n2} - i_{n3} + i_{n4}$$

$$S_{\lambda I_0} = \frac{\partial I_0}{\partial \lambda} (g_{m1} + g_{m2} + g_{m3} + g_{m4})$$

$$S_{\lambda I_0} = \frac{16 \times 7}{3} (g_{mn} + g_{mp})$$

M_0 and M_{00} do not contribute

ch)

$$g_{m2} \left(\frac{v_n + v_x}{2} \right)$$

$$= g_{m1} \left(\frac{v_n - v_x}{2} \right)$$

$$\therefore v_x = 0$$

$$I_0 = g_{m4} v_g + g_{m2} \times \frac{v_n}{2}$$

$$I_0 = \frac{v_n, in}{2} \left(\frac{g_{m4} g_{m1} + g_{m2}}{g_{m3}} \right) = 2g_{mn} \times \frac{v_n, in}{2}$$

$$\therefore S_{\lambda I_0} = g_{mn}^2 \times S_{v_n, in}$$

$$\therefore S_{v_n, in} = \frac{S_{\lambda I_0}}{g_{mn}^2} = \frac{16 \times 7}{3} \left(\frac{g_{mn} + g_{mp}}{g_{mn}^2} \right)$$

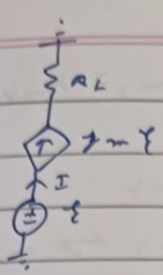
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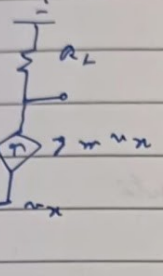
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A3.1a)



$$I = g_m v_x \Rightarrow 2i_x = \frac{1}{g_m} = a_s$$

$$\therefore g_m = \frac{1}{a_s}$$

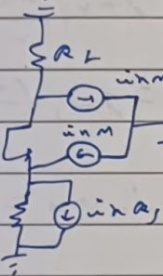


$$\bar{v} = \frac{v_s}{2a_s} = g_m v_x$$

$$\therefore v_o = +a_L \times \frac{v_s}{2a_s}$$

$$\therefore \text{gain} = a_L / 2a_s$$

b)



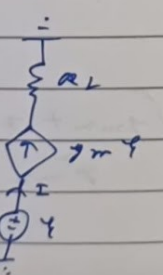
$$v_o = -\frac{i_{xm} \times a_L}{2} \oplus -i_{xa_s} \times a_L$$

$$\therefore S_{v_{no}} = a_L^2 \left(\frac{2 \times 7}{3 a_s} + \frac{4 \times 7}{a_s} \right)$$

$$S_{v_{no}} = 14/3 \times 7 a_L^2 / a_s$$

$$S_{v_{no}} = \frac{a_L^2}{4 a_s^2} \times S_{v_{in}} \Rightarrow S_{v_{in}} = \frac{56 \times 7 a_s}{3}$$

1)



$$I = g_m v_x$$

$$\therefore 2i_x = \frac{1}{g_m} = a_s$$

$$g_m = \frac{1}{a_s}$$

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$$v_{ox} = \frac{v_s \times R_L}{2R_s} \quad \left\{ \quad v_{on} = -\frac{v_s \times R_L}{2R_s} \right.$$

$$\therefore v_{os} = \frac{v_s R_L}{R_s} \Rightarrow G_m = \frac{R_L}{R_s}$$

d)

$$v_{ox} = -R_L (\bar{i}_{nM_1} + \bar{i}_{nR_s}) \quad \left\{ \quad v_{on} = -\frac{R_L}{N} \frac{\bar{i}_{nM_1} \times R_s}{2} \right.$$

$$v_{os} = -R_L (\bar{i}_{nR_s} - \frac{\bar{i}_{nM_2}}{N})$$

$$\therefore S_{V_{os}} = R_L^2 \times \frac{4kT}{R_s} \left(1 + \frac{2}{3N} \right) \quad (\because g_{m2} = N/R_s)$$

$$S_{V_{os}} = \frac{R_L^2}{R_s^2} \times S_{V_{n, in}} \Rightarrow S_{V_{n, in}} = \frac{4kT R_s (1 + \frac{2}{3N})}{3N}$$

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d)

$I + g_m(v_x - v_s) = v_s g_d$
 $\therefore I = v_s \left[g_d + \frac{(g_d + g_s) g_m}{g_d + g_s + g_m} \right]$
 $\therefore g_d + \frac{(g_d + g_s) g_m}{g_d + g_s + g_m} = \frac{1}{R_s} = g_s$
 $\therefore g_m = \frac{(g_d + g_s)(g_s - g_d)}{2g_d} = \frac{g_s^2 - g_d^2}{2g_d}$
 $g_m \rightarrow \infty \text{ for } g_d \rightarrow 0$

$v_x(g_d + g_s)$
 $= g_m(v_s - v_x)$
 $\therefore v_x = \frac{g_m v_s}{g_m + g_d + g_s}$

→

$(v_s - v_x) g_s = v_x g_d + 2g_m v_x$
 $\therefore v_x = \frac{v_s g_d}{g_s + g_d + 2g_m}$

$v_{ox} = 2g_m v_x R_L \quad \begin{cases} v_{on} = -2g_m v_x R_L \\ v_o = v_{ox} - v_{on} \end{cases}$

→ Gate voltages are negatives of each other by

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$$v_o = \frac{4g_m v_c g_d R_L}{g_s + g_d + 2g_m} \Rightarrow \text{Gain} = \frac{2g_m g_d R_L}{g_s + g_d + 2g_m}$$

$$\text{Gain} = \frac{R_L (g_s - g_d) \frac{g_d}{g_d}}{1 + (g_s - g_d)/g_d}$$

$$\text{Gain} = R_L (g_s - g_d) = R_L \left(\frac{1}{R_s} - \frac{1}{R_d} \right) = \frac{R_L}{R_s}$$

cf)

$$v_{o1} = -R_L [i_{nm} + g_m (v_{cm} - v_d)]$$

$$v_{o2} = -R_L [i'_{nm} + g_m (v_d - v_{cm})]$$

$$i_{ns} + g_m (v_{cm} - v_d) + i_{nm} = v_d (g_s + g_d) \quad \text{--- (1)}$$

$$i'_{ns} + g_m (v_d - v_{cm}) + i'_{nm} = v_{cm} (g_s + g_d) \quad \text{--- (2)}$$

$$v_o = v_{o1} - v_{o2} = R_L [2g_m (v_d - v_{cm}) + i'_{nm} - i_{nm}] \quad \text{--- (3)}$$

Solving (1), (2) and (3):

$$v_o = \frac{R_L (g_s + g_d) (i'_{nm} - i_{nm}) + 2g_m (i_{ns} - i'_{ns})}{(2g_m + g_s + g_d)}$$

$$S_{vt} = \frac{R_L^2}{(2g_m + g_s + g_d)^2} \left[\frac{16kTg_m (g_s + g_d)^2}{3} + 32g_m^2 \frac{kT}{R_s} \right]$$

$\therefore S_{vt} \rightarrow 8kT R_L^2 / R_s$ when $g_m \rightarrow \infty, g_d \rightarrow 0$

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$$v_{n\phi} = 2\alpha_L(z_s - z_h) \times v_{n, in}$$

$$S_{v_{n\phi}} = 4\alpha_L^2 (z_s - z_h)^2 \times S_{v_{n, in}}$$

$$\therefore S_{v_{n, in}} = \frac{4\pi\gamma [2z_m^2 z_s + \frac{1}{3}(z_s + z_h)^2 z_m]}{(z_s - z_h)^2 (2z_m + z_s + z_h)^2}$$

Substituting $z_m = \frac{z_s^2 - z_h^2}{2z_h}$

$$S_{v_{n, in}} = \frac{2\pi\gamma}{3(z_s - z_h) \times z_s^2} [3z_s^2 - 2z_s z_h + z_h^2]$$

$$S_{v_{n, in}} = 2\pi\gamma \alpha_s \quad \text{for } z_h \rightarrow 0$$